Valgrind

Computer Programming 1

2026-10-30

Outline

Agenda

- What is Valgrind?
- The Buggy C++ Program
- Running Memcheck
- Reading the Output & Fixing Bugs

What is Valgrind?

- A dynamic analysis framework for building profiling and debugging tools.
- It detects memory management and heap bugs in C and C++.
- Catch memory leaks, buffer overruns, and invalid deletions.
- Works directly on compiled binaries (compile with `-g` for source line numbers).

The Buggy Code

Source File: buggy.cpp

```
#include <iostream>

void leaky_function() {
  int *arr = new int[10];
  arr[10] = 42;
}

void lazy_function(int *arr) {
  arr[3] = 42;
}

int main() {
  int * a;
  std::cout << "Starting program..." << std::endl;
  leaky_function();
  lazy_function(a);
  std::cout << "Done!" << std::endl;
  delete a[];
```

Running Valgrind

Compile
Compile with debugging symbols (`-g`):

```
g++ -g buggy.cpp -o buggy
```

Run
Run the binary through Valgrind:

```
valgrind ./buggy
```

Reading the Output & Fixing Bugs

Out-of-Bounds Write Output

```
==56789== Invalid write of size 4
==56789==    at 0x40120A: leaky_function() (buggy.cpp:8)
==56789==    by 0x40124B: main (buggy.cpp:15)
==56789== Address 0x4a43048 is 0 bytes after a block of size 40 alloc'd
```

Memory Leak Summary

```
==56789== HEAP SUMMARY:
==56789==     in use at exit: 40 bytes in 1 blocks
==56789==   definitely lost: 40 bytes in 1 blocks
```